EAM 2023

Reliability for multilevel data: A correlation approach

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13-07-2023

Reliability and Measurement Error

- A **reliable** method or measurement instrument is one which gives the same or similar results when it is repeated.

- However, few studies define or report reliability specifically for different hierarchical levels in multilevel data (Geldhof et al., 2014, ten Hove et al., 2022).

- Two recent papers (ten Hove et al., 2022; Schönbrodt et al., 2022) address how to quantify reliability in multilevel settings based on generalizability theory.
  - Nevertheless, both approaches give **inconsistent results** regarding their definition of cluster-level reliability.

Classical Test Theory

An observed-score Y_{ij} for subject i at occasion j can be decomposed into two parts: a true score T_{ij} and a random measurement error ϵ_{ij}

$$Y_{ij} = T_{ij} + \epsilon_{ij}$$

- where $E[Y_{ij}] = T_{ij}, Cor(\epsilon_{ij}, \epsilon_{ij'}) = Cor(\epsilon_{ij}, T_{ij}) = 0$

The reliability is defined as

$$Cor(Y_{ij}, T_{ij})^2 = \frac{\sigma_T^2}{\sigma_Y^2} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

- It is a ratio of the true-score variance σ_T^2 over the total variance σ_Y^2.

Classical Test Theory

Under the parallel assumptions

- $E[Y_{ij}] = E[Y_{ij'}] = T_i$, $Var(Y_{ij}) = Var(Y_{ij'})$
- The reliability coefficient is also equal to the correlation between two replicates Y_{ij} and $Y_{ij'}$, i.e.,

$$Cor(Y_{ij}, Y_{ij'}) = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, T_{ij})^2.$$

Generalizability Theory

Extending classical test theory

- Recognize and estimate the magnitude of the multiple sources of variation (Cronbach, 1972; Brennan, 2001).
  - such as subjects, raters, items, occasions, etc.
- E.g., a three-way $Subject(S_i) \times Rater(R_j) \times Day(D_t)$ design

	Rater 1		Rater 2	
Subject	Day 1	Day 7	Day 1	Day 7
1	Y_{111}	Y_{112}	Y_{121}	Y_{122}
2	Y_{211}	Y_{212}	Y_{221}	Y_{222}
3	Y_{311}	Y_{312}	Y_{321}	Y_{322}
.	.	.	.	.
.	.	.	.	.
10	Y_{1011}	Y_{1012}	Y_{1021}	Y_{1022}

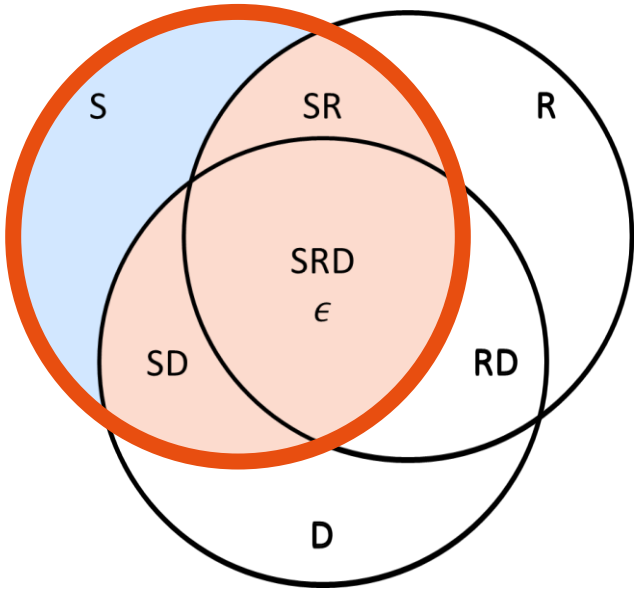
$$Y_{ijt} = \mu + S_i + R_j + D_t + (SR)_{ij} + (SD)_{it} + (RD)_{jt} + \epsilon_{ijt}$$

Generalizability Theory

Relative reliability (R_{rel})

- Consistency between subjects' rank-orderings

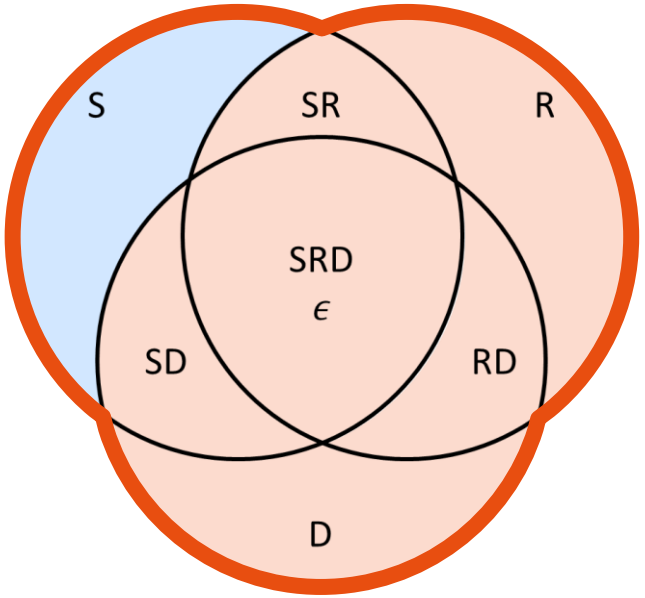
$$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$$



Absolute reliability (R_{abs})

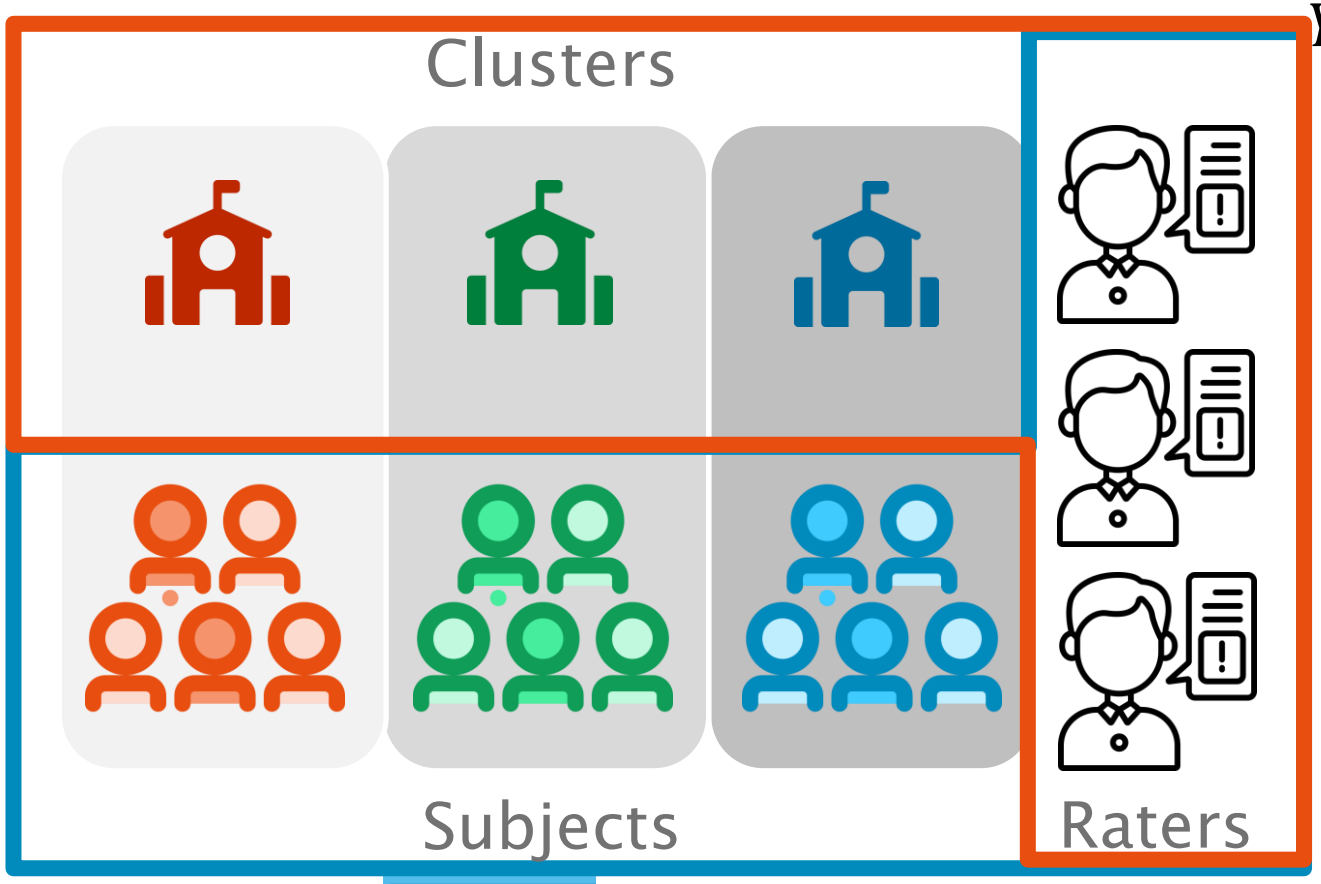
- Agreement between subjects' absolute scores

$$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$$

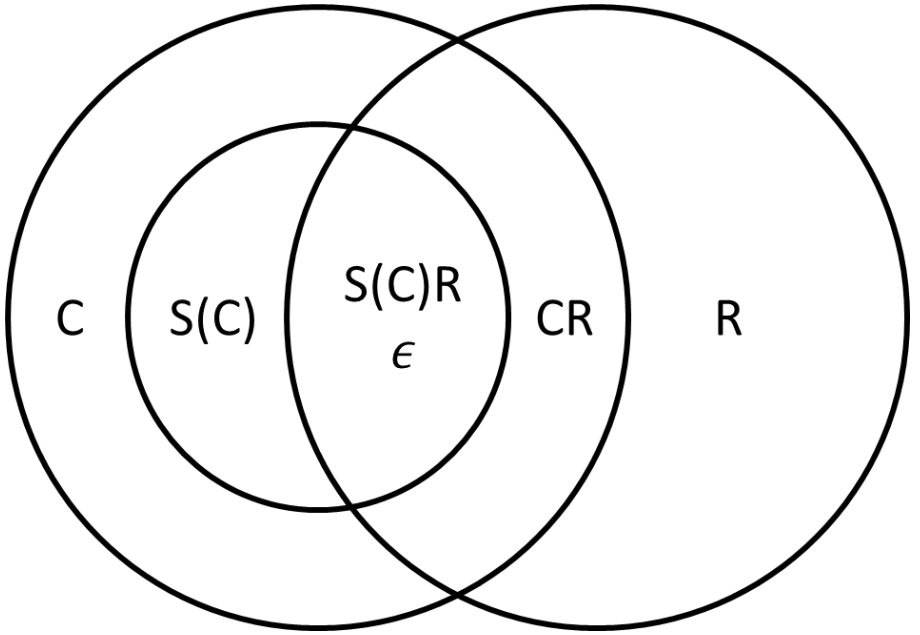


Nested Structures

E.g., subjects ($S_{i(k)}$) are nested within clusters (C_k), and raters (R_j) are crossed with S & C.



$$Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$$



Reliability for the different levels?

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Interrater Reliability for Multilevel Data: A Generalizability Theory Approach

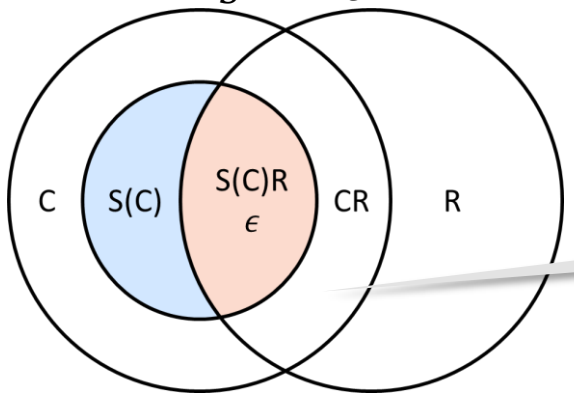
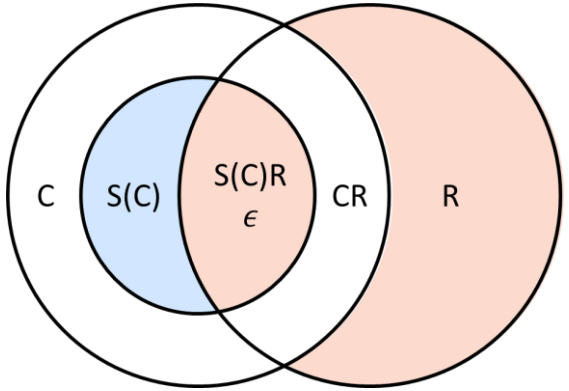
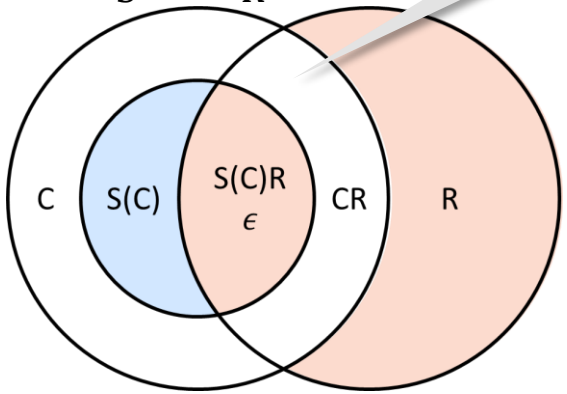
Debby ten Hove, Terrence D. Jorgensen, and L. Andries van der Ark
Research Institute of Child Development and Education, University of Amsterdam

Measuring motivational relationship processes in experience sampling: A reliability model for moments, days, and persons nested in couples

Behavior Research Methods (2022) 54:1869–1888
https://doi.org/10.3758/s13428-021-01701-7

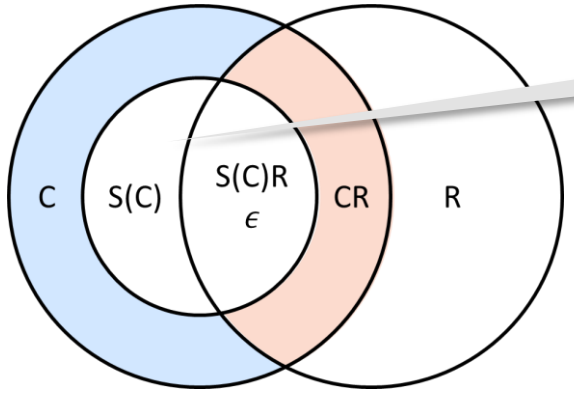
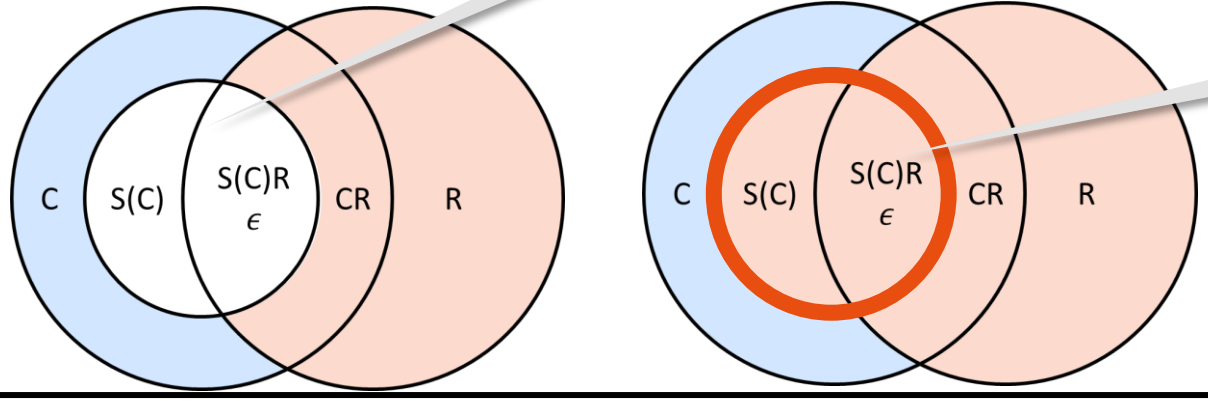
Felix D. Schönbrodt[1] · Caroline Zygar-Hoffmann[1] · Steffen Nestler[2] · Sebastian Pusch[3] · Birk Hagemeyer[3]

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	(not defined)
R^S_{rel}		
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 

Ignore cluster-level components

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Cluster-level	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$	(not defined)
	R^C_{rel} 	
	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
	R^C_{abs} 	

Ignore subject-level components

Include everything

Comparison

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Subject level	Do not contain any cluster-related (higher level) variance components (i.e., σ_C^2, σ_{CR}^2)	
Cluster level	Do not contain any subject-related variance (lower level) components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$)	Contain subject-related variance components
Types	Separate **relative** and **absolute** reliability	Only discuss **absolute** reliability

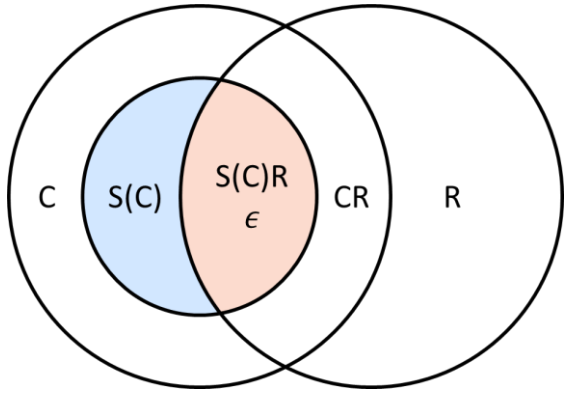
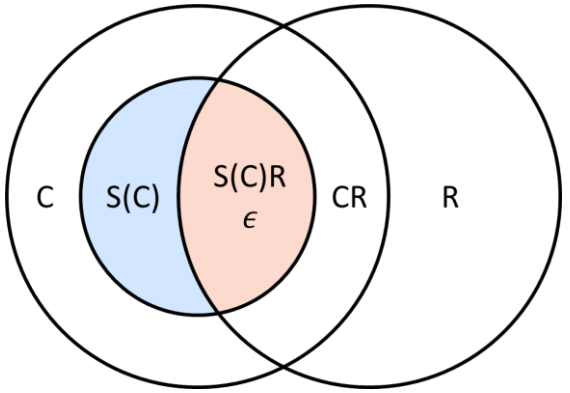
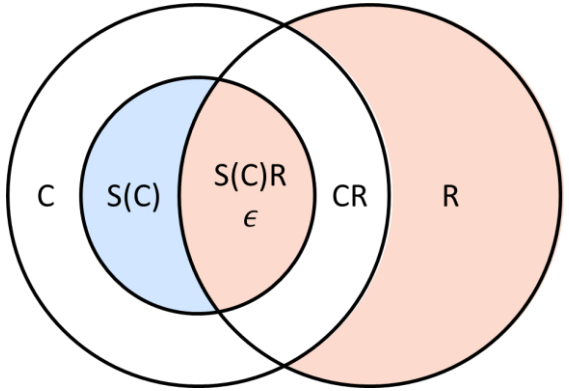
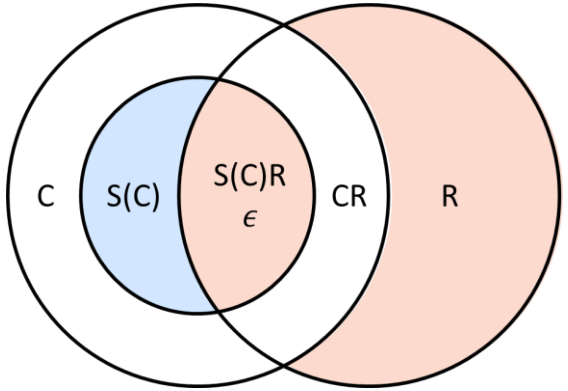
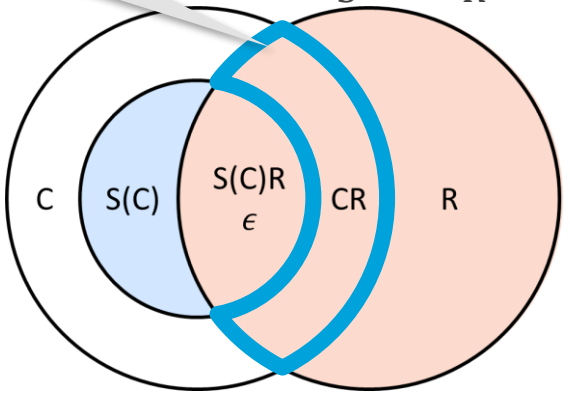
Correlation Approach for Multilevel Data

The reliability coefficient can be defined as the **correlation** between two measurements from the same subject i on different raters (or occasions, etc.) j and j' (Vangeneugden et al., 2005).

	Subject-level	Cluster-level
Relative reliability	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j')$	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j')$
Absolute reliability	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k)$	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k))$

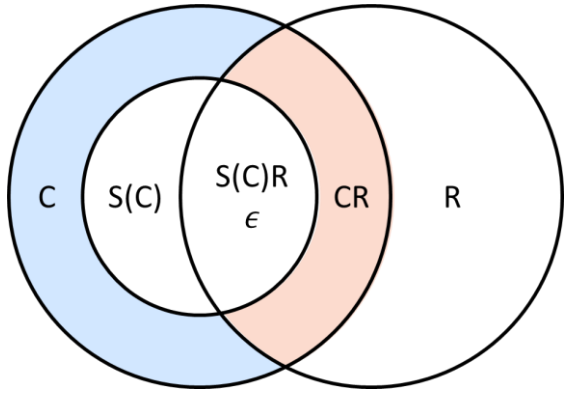
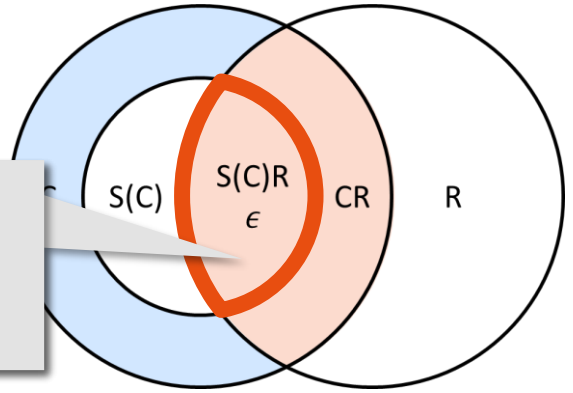
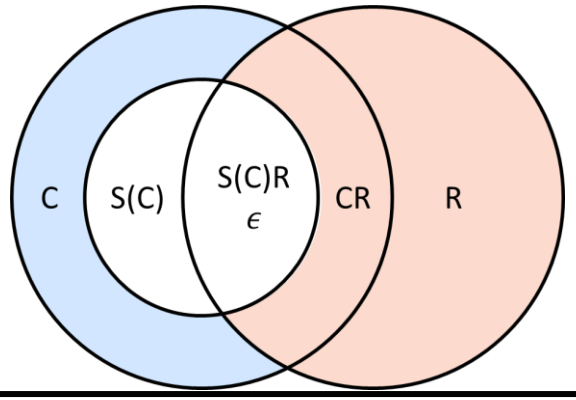
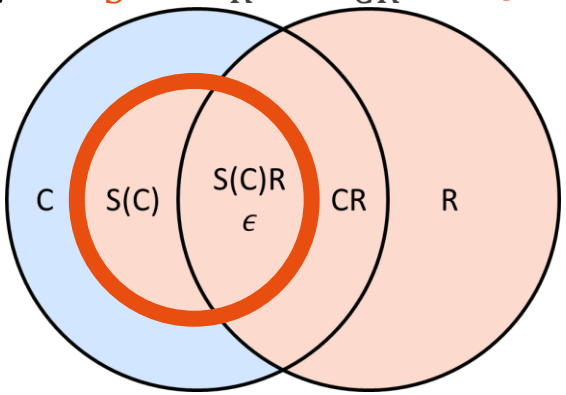
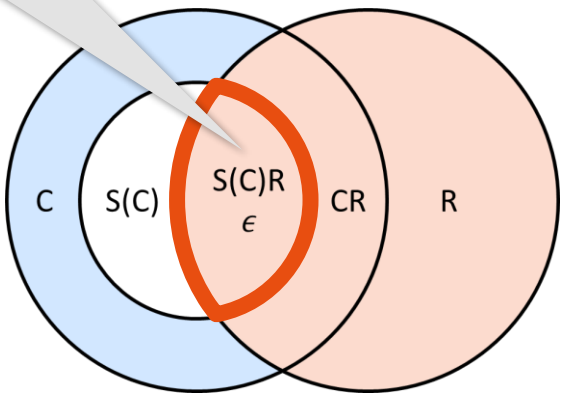
$C_k, R_j, R_{j'}, (CR)_{kj}, (CR)_{kj'}$

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	(not defined)	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}			
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$ 

Because (CR) is random and influences rater effects

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + (CR)_{kj} + \epsilon_{i(k)j}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Cluster-level	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$	(not defined)	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j'\right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
	R^C_{rel} 		
	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k)\right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
	R^C_{abs} 		

With measurement error variance

Discussion and Future Work

- Point out the different reliability definitions for multilevel data in the latest literature (ten Hove et al., 2022; Schönbrodt et al., 2022).
  - there is no consensus on defining the reliability coefficient
- Using correlation is a promising approach to define reliability for multilevel settings.
  - a principal rule to derive our formula
  - easy to understand and has the flexibility for complicated models, e.g., (generalized) linear mixed-effects models (Molenberghs et al. 2007)
- Challenges
  - reliability coefficients for linear mixed-effects models are inconsistencies between ICC, GT, and the correlation approach
  - the meaning of conditioning

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Q&A

Reliability for multilevel data: A correlation approach

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Reliability for other nested structures lists in the appendix slides

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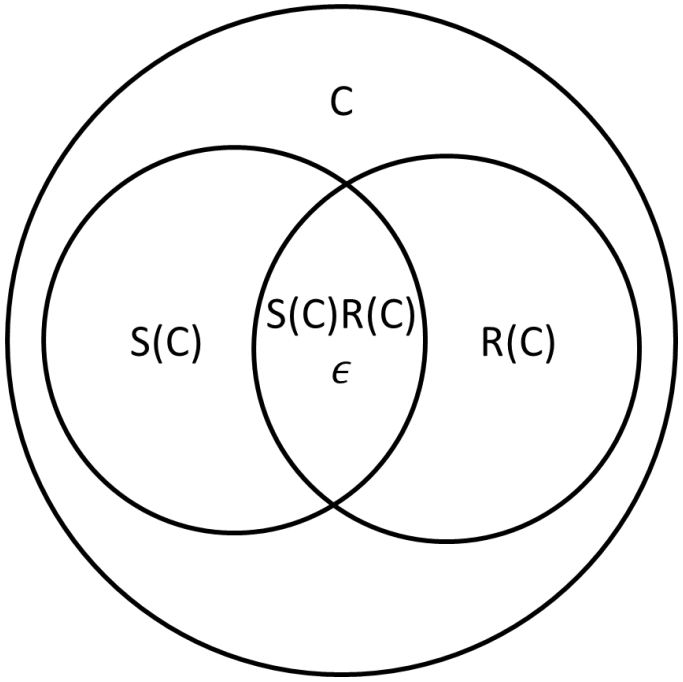
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Other Nested Structure Examples

Subjects (S) are nested within clusters (C), and …

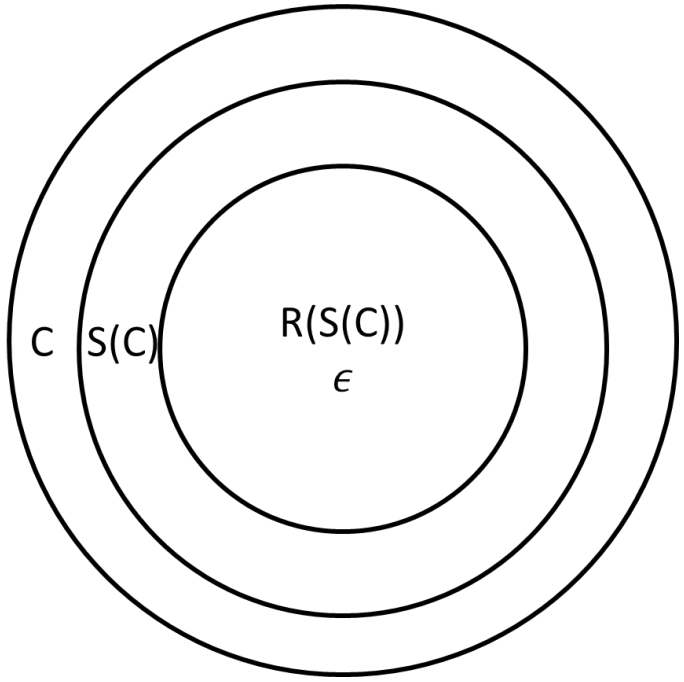
- Case 2: raters (R) are *nested within* C and *crossed with* S

$$Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$$

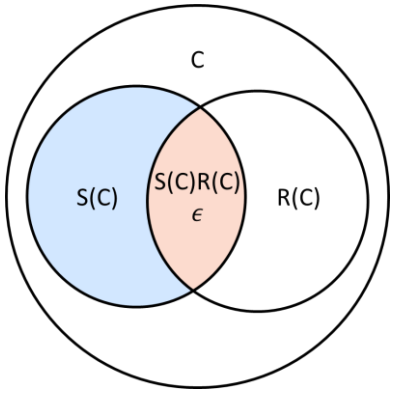
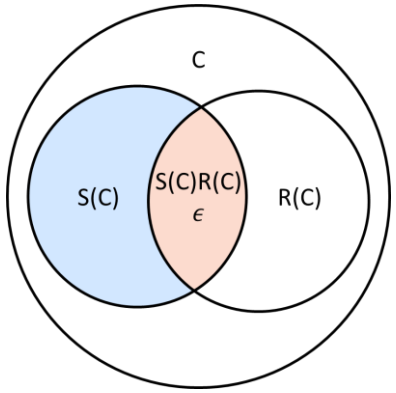
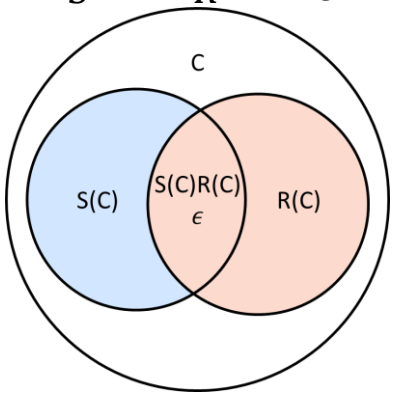
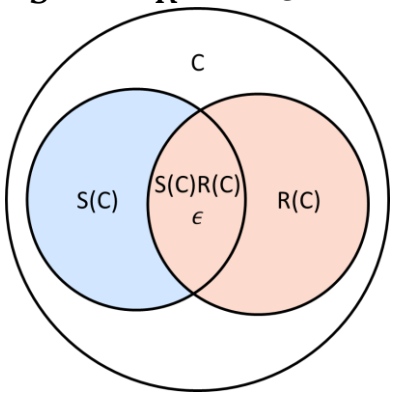
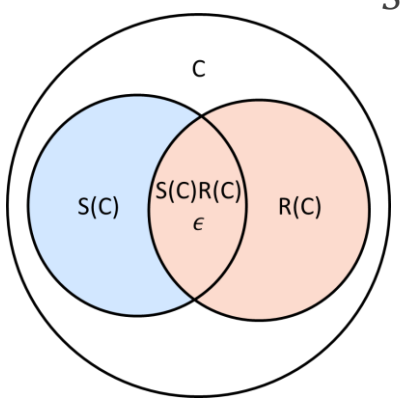


- Case 3: raters (R) are *nested within* S & C

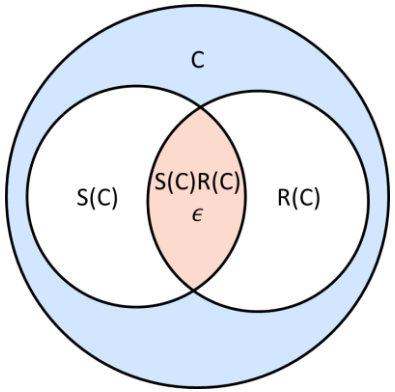
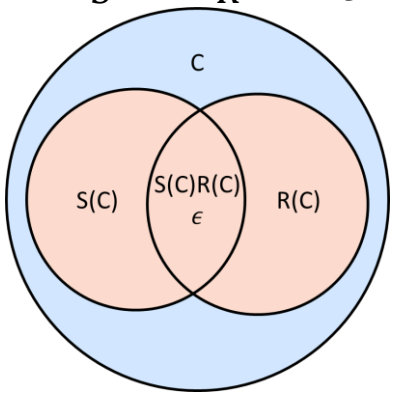
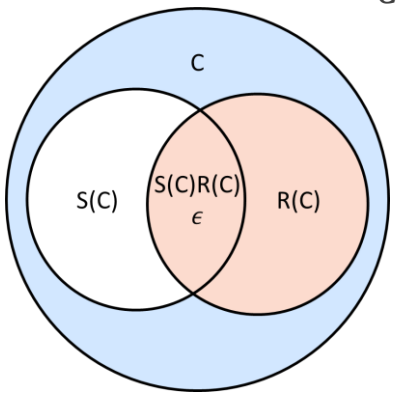
$$Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$$



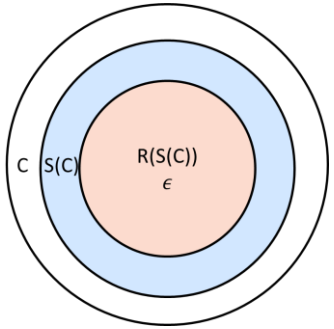
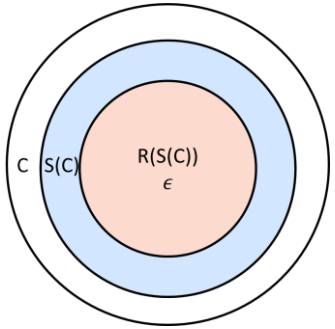
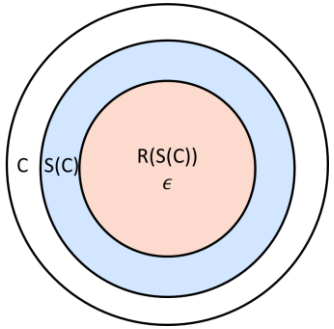
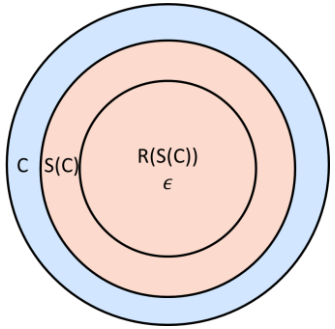
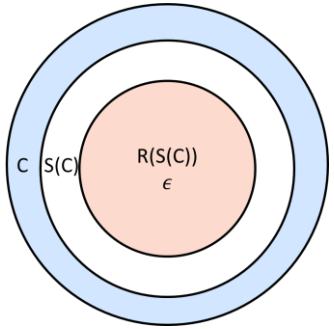
Case 2: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$		$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k, j, j' \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}		(not defined)	
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 

Case 2: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Cluster-level			$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i, j, j' \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$
R_{rel}^{C}	Cannot be defined because it ignores all lower-level variances	(not defined)	
R_{abs}^{C}		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 

Case 3: $Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)*	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S			
Cluster-level		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$
R^C	Cannot be defined because it ignores all lower-level variances		

* See also Raudenbush et al. (1991), Cranford et al. (2006), and Shrout & Lane (2012).